

AI-Powered Civic Cleanliness & Waste Intelligence Operating System

A Closed-Loop Urban Cleanliness Platform Connecting Citizens, Field Sanitation Squads, and Municipal Authorities with Deep Learning Computer Vision, Automated SLA Enforcement, and GIS Spatial Intelligence.

Initiative: Smart India Hackathon (SIH) Technical Submission

Architecture: 3-Tier Enterprise Web Platform • Modular MVC Services • Edge AI Vision

Primary Stack: React 19, Vite, Tailwind CSS, Node.js, Express.js, MongoDB Geospatial, YOLOv8, Socket.io

Regulatory Compliance: CPCB & WHO Biomedical Waste Management Rules, Smart Cities Mission Guidelines

1. Executive Summary & Urban Sanitation Context

Municipal solid waste management in rapidly expanding urban centers faces profound structural challenges: **(1)** Unstructured, vague civic complaints lacking actionable geolocation coordinates or material categorization; **(2)** Dangerous co-mingling of infectious biomedical waste (syringes, needles, contaminated swabs) with general dry municipal garbage; **(3)** Redundant dispatches of multiple sanitation crews to identical locations; **(4)** Absence of verifiable ground-level post-cleanup audits; and **(5)** Unmonitored, unenforced Service Level Agreements (SLAs) resulting in administrative inertia.

CleanTrack introduces an automated, closed-loop urban cleanliness operating system that addresses these challenges. Citizens report waste through an AI-guided interface with automated GPS geolocation. CleanTrack's dual **YOLOv8 Computer Vision Ensemble** performs instantaneous neural inference to classify waste categories, compute exact camera frame coverage percentages, segregate biohazards according to **CPCB & WHO statutory rules**, and assign an automated 5-factor priority score. Unresolved grievances are escalated automatically by a background SLA cron worker, while field cleanup authenticity is verified through an interactive photographic comparison slider rewarding citizens with Green Points.

The Closed-Loop Cleanliness Lifecycle:

- [1. Citizen Capture]** Live camera photo + GPS coordinate pinpoint →
- [2. AI Vision Scan]** Dual YOLOv8 polymer & clinical sharps detection →
- [3. Intelligent Triage]** 5-factor mathematical scoring & SLA deadline calculation →
- [4. Squad Dispatch]** Specialized team assignment (Plastic Compactor / Biohazard Sharps) →
- [5. Field Cleanup]** Field sanitation squad uploads 'After' photographic proof →
- [6. Citizen Verification]** Citizen validates cleanup via interactive split-view slider →
- [7. Green Points Ledger]** Gamified citizen rewards & municipal performance index update.

2. Comprehensive Module Breakdown

CleanTrack is architected into 10 decoupled, highly cohesive modules spanning citizen reporting, computer vision inference, operational triage, municipal governance, and macro-level urban analytics.

Module 1: Citizen Engagement & Smart Complaint Reporting

Functionality & Capabilities: Enables citizens to lodge actionable civic grievances in under 30 seconds. Features browser-native GPS coordinate capture, reverse geocoding to municipal ward numbers and landmarks, local image selection/camera capture, real-time client-side pre-scan visualizers, and a live complaint tracker tracing status from submission to citizen sign-off.

Technologies Employed: React 19, React Router DOM 7, HTML5 Geolocation API, Leaflet GIS, Lucide Icons, Multer, Express.js REST API

Module 2: AI Computer Vision & Deep Learning Waste Classification Engine

Functionality & Capabilities: A multi-model deep learning inference subsystem. Orchestrates two fine-tuned YOLOv8 convolutional neural networks (plastic_best.pt and biomedical_best.pt). Identifies 25+ waste classes (PET, HDPE containers, LDPE films, PP rigids, syringes, hypodermic needles, surgical gloves, soiled bandages, test tubes), performs Non-Maximum Suppression (NMS) and IoU de-overlapping, calculates exact surface area coverage %, assigns WHO/CPCB segregation streams, and renders annotated visual bounding boxes.

Technologies Employed: Ultralytics YOLOv8, PyTorch, Python 3.10+, OpenCV (cv2), Pillow (PIL), NumPy, Node.js Child Process IPC

Module 3: Intelligent Triage, Priority Scoring & Dynamic SLA Escalation

Functionality & Capabilities: Eliminates manual grievance prioritization. Evaluates incoming complaints against a 5-factor scoring model: Waste Hazard (30%), Visual Extent (30%), Location Sensitivity (25%), Chronic Recurrence (10%), and Time Elapsed (5%). Computes a 0-100 score and assigns strict SLA resolution deadlines (Critical = 12h, High/Medium = 24h, Normal = 48h). A background cron engine automatically escalates overdue tickets to CRITICAL, appends priority history, records an audit log, and pushes emergency alarms.

Technologies Employed: node-cron, Node.js, Express Services, MongoDB Mongoose Schemas, Socket.io Real-Time Broadcast

Module 4: GIS Spatial Mapping, Duplicate Clustering & Hotspot Analytics

Functionality & Capabilities: Utilizes MongoDB 2dsphere spherical geometry indexes to identify duplicate complaints within a 50-meter radius, preventing redundant crew dispatches. Aggregates complaint spatial coordinates to detect chronic waste hotspots (>=3 reports in 14 days) and renders interactive color-coded ward heatmaps.

Technologies Employed: Leaflet.js, React-Leaflet, MongoDB Geospatial Indexing (\$geoNear, 2dsphere), Tailwind CSS

Module 5: Municipal Operations, Field Dispatch & Squad Assignment

Functionality & Capabilities: Operational command center for municipal zonal triage officers. Features a live Priority Queue sorted by SLA countdown, a Quick-Assign Dispatch Modal to allocate specialized response teams (Squad Alpha: Plastic Compactor, Squad Bravo: Biohazard Sharps Unit, Squad Charlie: Organic Composting), and squad instruction broadcasting.

Technologies Employed: React 19 Context API, Mongoose Relational Population (ObjectId), Express.js REST APIs

Module 6: Civic Cleanup Verification & Before/After Visual Audit

Functionality & Capabilities: Guarantees ground-level cleanup authenticity. Sanitation crews must upload an on-site 'After' photograph to complete an assignment. Citizens interact with a side-by-side Before/After comparison slider to validate or reject the cleanup. Verified cleanups automatically credit +50 Green Points and trigger celebratory animations.

Technologies Employed: Custom React Touch/Mouse Drag Slider, Express Verification Controller, canvas-confetti

Module 7: Gamification, Green Points Ledger & Citizen Rewards Center

Functionality & Capabilities: Incentivizes sustained civic participation. Citizens earn points for legitimate reports (+25 pts) and confirmed cleanups (+50 pts). Features 5 progression tiers (Eco Rookie to Clean City Champion), dynamic progress bars, and an immutable financial-grade transaction ledger for redeeming public transport vouchers and utility rebates.

Technologies Employed: MongoDB Reward & Transaction Schema, React UI Components, Custom Gamification Engine

Module 8: Administrative Governance, Municipal Watch & RBAC Security

Functionality & Capabilities: Empowers city municipal commissioners with high-level governance. Enforces strict Role-Based Access Control (RBAC) across Citizen, Municipal Staff, and Administrator roles. The Municipal Watch page monitors ward-by-ward Mean Time to Resolve (MTTR), squad workload, staff KPI leaderboards, and immutable system audit logs.

Technologies Employed: JSON Web Tokens (JWT), Bcrypt.js Password Hashing, Express Auth & Role Middleware

Module 9: Real-Time Event Dispatch & In-App Notification Hub

Functionality & Capabilities: Bi-directional event streaming infrastructure. Dispatches instantaneous push notifications for ticket assignments, SLA breach warnings, cleanup confirmations, and reward credits without page refreshes. Persists historical notifications with read/unread statuses.

Technologies Employed: [Socket.io \(WebSocket Client & Server\)](#), [Notification Model](#), [React Notification Context](#)

Module 10: Executive Analytics, Telemetry & Clean City Pulse Engine

Functionality & Capabilities: Macro-level urban intelligence dashboard. Calculates the Clean City Pulse Score (0-100 composite index) based on resolution speed, SLA compliance, and open hotspot density. Generates interactive visual breakdowns of monthly complaint trends, waste type distributions, and AI model inference latency telemetry.

Technologies Employed: [Recharts 3 SVG Charting Library](#), [MongoDB Aggregation Pipelines \(\\$group, \\$match, \\$facet\)](#)

3. Complete Technology Stack Matrix

CleanTrack utilizes a modern, production-grade technology ecosystem engineered for low-latency responsiveness, high concurrency, strict security, and real-time computer vision inference.

Domain / Layer	Technology	Version	Role & Architectural Function in CleanTrack
Frontend Framework	React.js	19.2.x	Component-based UI architecture, virtual DOM reconciliation, state reactivity, and multi-role workspace management
Build Engine	Vite	8.2.x	Next-gen ESM development server, sub-second Hot Module Replacement (HMR), and optimized production bundling
UI Styling	Tailwind CSS	3.4.x	Utility-first responsive styling, glassmorphism card surfaces, and color-coded CPCB waste stream design tokens
Client Routing	React Router DOM	7.18.x	Declarative SPA routing, protected route guards, role barrier redirects, and layout encapsulation
Geospatial GIS	Leaflet & React-Leaflet	1.9.4 / 5.0	Interactive city ward maps, draggable coordinate pin selectors, and color-coded hotspot clusters
Data Visualization	Recharts	3.10.x	Composable SVG charts, complaint trend curves, category distributions, and Clean City Pulse gauges
Vector Icons & FX	Lucide React & Confetti	Latest	Lightweight vector icons and celebratory canvas particle animations upon cleanup validation
Backend Runtime	Node.js	v20+ / v22	Asynchronous, event-driven JavaScript server runtime for high-throughput non-blocking operations
Web Server Framework	Express.js	4.21.x	Modular RESTful API routing, MVC controllers, middleware chaining, and centralized error handling
Database Layer	MongoDB	8.x / Atlas	Document-oriented NoSQL storage with native GeoJSON spherical geometry indexing for spatial queries
Data Modeling (ODM)	Mongoose	8.10.x	Schema validation, strict typings, 2dsphere indexes, pre-save hooks, and ObjectId relational population
Real-Time WebSockets	Socket.io	4.8.x	Bi-directional, event-driven WebSocket communication for instantaneous live alarms and status sync
Background Cron Engine	Node-Cron	3.0.x	Background task scheduler continuously evaluating and escalating breached SLA complaint deadlines
AI / ML Framework	Ultralytics YOLOv8	8.x	Dual custom neural network models for real-time plastic and biomedical waste detection
Deep Learning Core	PyTorch & Python	3.10+ / 2.x	Neural network tensor execution, model weight processing (best.pt), and GPU-accelerated inference
Image Processing	OpenCV, PIL, NumPy	Latest	Image decoding, bounding box rendering, IoU duplicate box suppression, and area math
Auth & Cryptography	JWT & Bcrypt.js	9.0.x / 2.4.x	Stateless token authorization and 10-round salted one-way password hashing
API Security	Helmet & Rate-Limit	Latest	HTTP security headers, XSS mitigation, and IP-based endpoint throttling against brute force
File Multipart Upload	Multer	1.4.x	Local disk storage and memory buffering of incoming photographic complaint evidence

4. In-Depth Architectural Deep Dives of Core Technologies

1. React 19 (Frontend Presentation Framework)

React 19 forms the core foundation of CleanTrack's user interface. It utilizes a declarative component model where UI elements are encapsulated into reusable units such as Complaint Cards, AI Scanners, and Before/After Sliders. React 19's optimized virtual DOM reconciliation ensures instantaneous UI updates when WebSocket events arrive. In CleanTrack, the frontend is organized into 4 distinct layout zones: Public Informational routes, Citizen Workspace, Municipal Staff Triage Center, and Executive Administration Dashboard.

2. Vite 8.2 (Next-Generation Frontend Build Engine)

Vite provides the development and build infrastructure for CleanTrack. Unlike traditional bundlers that pre-bundle entire applications before serving, Vite serves source code over native browser ES Modules (ESM), providing sub-second dev server start times and instantaneous Hot Module Replacement (HMR). During production compilation, Vite leverages Rollup to tree-shake unused code and output highly optimized, cache-busted static assets.

3. Tailwind CSS 3.4 (Utility-First Responsive Styling)

Tailwind CSS is an atomic CSS framework that eliminates bulky custom stylesheets by providing utility classes directly within JSX. CleanTrack leverages Tailwind to create a unified design system with semantic color tokens corresponding to CPCB waste streams (Yellow for biohazard incineration, Red for autoclave plastics, Blue for glassware, Amber for SLA alerts). Tailwind's responsive breakpoints ensure full operational usability across desktop monitors, field tablets, and citizen smartphones.

4. Leaflet.js & React-Leaflet (Geospatial GIS Engine)

Leaflet is a lightweight open-source JavaScript library for interactive mapping. React-Leaflet provides idiomatic React bindings for Leaflet's map instances. In CleanTrack, Leaflet renders OpenStreetMap tile layers, custom SVG markers with animated severity halos, interactive radius overlays, and draggable location pickers with automated latitude/longitude binding.

5. Recharts 3.10 (Declarative SVG Data Visualization)

Recharts is a charting library built natively on React and D3.js, rendering pure Scalable Vector Graphics (SVG). CleanTrack employs Recharts to visualize civic metrics: weekly complaint volume trends, ward-by-ward resolution velocities, waste composition breakdowns, and the composite Clean City Pulse radial gauge.

6. Node.js & Express.js 4.21 (Backend REST & Services Architecture)

Node.js executes JavaScript on Chrome's V8 engine using an asynchronous, non-blocking I/O event loop. Express.js organizes CleanTrack into a structured 3-tier architecture: (1) HTTP Routes, (2) Controllers executing business validation, and (3) Reusable Services (aiService, escalationService, rewardService, notificationService). This ensures complete separation of concerns and maintainability.

7. MongoDB 8 & Mongoose 8.10 (Geospatial NoSQL Database & ODM)

MongoDB stores records as flexible JSON-like BSON documents. Mongoose enforces strict schema typings, validation rules, and relational ObjectIds. CleanTrack implements a **2dsphere geospatial index** on the Complaint collection, enabling sub-millisecond execution of spherical geometry queries (\$geoNear with maxDistance: 50m) to immediately detect and cluster duplicate complaints.

8. Socket.io 4.8 (Bi-Directional Real-Time WebSocket Infrastructure)

Socket.io establishes persistent, low-latency WebSocket connections between web clients and the backend server. CleanTrack uses Socket.io to broadcast live events across dedicated rooms. When a citizen submits a critical complaint or an SLA breaches, municipal staff dashboards update in real time with acoustic and visual alerts without requiring manual page refreshes.

9. Node-Cron 3.0 (Automated SLA Background Engine)

Node-Cron executes scheduled cron jobs inside the Node.js process using standard POSIX crontab syntax. CleanTrack runs an automated SLA watchdog every 60 seconds (*/* * * * *). The job checks for unresolved complaints where dueAt < NOW, automatically elevates their priority to CRITICAL, logs an immutable audit entry, and broadcasts emergency escalation alerts.

10. Ultralytics YOLOv8 & PyTorch (Computer Vision & Deep Learning)

YOLOv8 (You Only Look Once v8) is a state-of-the-art Single-Shot Object Detector that processes images in a single forward pass. CleanTrack orchestrates a dual-model ensemble: (1) A specialized **Plastic Polymer Model** trained on diverse PET, HDPE, LDPE, and composite packaging; and (2) A **Biomedical Waste Model** trained on WHO/CPCB medical classes (syringes, needles, gauze, gloves, test tubes). The Python inference engine executes Non-Maximum Suppression (NMS), computes area coverage fractions, and yields structured JSON telemetry alongside annotated visual bounding boxes in ~140ms.

11. OpenCV, Pillow (PIL) & NumPy (Computer Vision Preprocessing & Math)

OpenCV and PIL handle on-the-fly image decoding, aspect-ratio scaling, and bounding box visualization. NumPy executes high-speed matrix operations to calculate Intersection-over-Union (IoU) matrices, suppressing overlapping duplicate bounding boxes and computing exact spatial coverage fractions across camera frames.

12. JWT & Bcrypt.js (Stateless Authentication & Cryptographic Security)

CleanTrack ensures robust stateless security using signed JSON Web Tokens (HMAC-SHA256). Passwords are encrypted using Bcrypt.js with an adaptive work factor of 10 salt rounds. Role-Based Access Control (RBAC) middleware intercepts every API route, ensuring citizens cannot access municipal dispatch queues and staff cannot modify administrative system configurations.

13. Helmet & Express-Rate-Limit (Web Application Protection)

Helmet configures critical HTTP response security headers (Content Security Policy, X-Frame-Options, X-XSS-Protection). Express-Rate-Limit enforces request limits on public endpoints to protect against brute-force attacks and Denial of Service (DoS).

14. Multer (Multipart Form Data & File Handling)

Multer is an Express middleware that handles multipart/form-data uploads. It buffers incoming photographic evidence, validates MIME types, and streams files directly to disk storage or cloud storage buckets.

5. Mathematical Models & Core Algorithms

1. Multi-Factor AI Priority Score Algorithm (0 to 100 Scale)

Every complaint is evaluated by an automated multi-parametric scoring formula that balances hazard danger with visual severity and community context:

Priority Score = W_type + V_extent + L_sensitivity + R_recurrence + T_time

- **Waste Type Hazard (W_type, Max 30 pts):** Syringes/Sharps = 30 pts; Bio-contaminated Cotton/Bandages = 22 pts; Plastic/Packaging = 12-18 pts; Clean = 0 pts.

- **Visual Area Coverage (V_extent, Max 30 pts):** Calculated directly from YOLO bounding box pixel dimensions: $\min(30, 18 + \text{Coverage\%} * 0.2 + \text{ItemCount} * 0.5)$.

- **Location Sensitivity (L_sensitivity, Max 25 pts):** Hospital/School zone = 25 pts; Commercial/Market = 20 pts; Residential = 15 pts; Open Suburban = 10 pts.

- **Chronic Recurrence Factor (R_recurrence, Max 10 pts):** Evaluates historical complaints within 100m radius over past 30 days.

- **Time Elapsed Aging (T_time, Max 5 pts):** Incremental priority boost as SLA deadline nears expiration.

2. Geospatial Duplicate Detection & Clean City Pulse Index

- **50-Meter Haversine Geospatial Duplicate Clustering:**

$\text{Distance}(P1, P2) = 2R * \arcsin(\sqrt{\sin^2(d_phi/2) + \cos(phi1) * \cos(phi2) * \sin^2(d_lambda/2)}) \leq 0.050 \text{ km}$

Executed natively inside MongoDB via \$geoNear with spherical geometry coordinates. If an active, unresolved complaint is discovered within 50m, the new submission is linked as a corroborating duplicate report.

- **Clean City Pulse Composite Formula (0 to 100 Index):**

Clean City Pulse = (Resolution_Rate * 0.45) + (SLA_Adherence * 0.35) + (Hotspot_Control * 0.20)

Provides city commissioners with an objective, real-time index of municipal sanitation health.

6. Database Architecture & Key Data Models

CleanTrack implements 10 relational Mongoose schemas in MongoDB designed for ACID transaction safety, relational consistency, and rapid indexed spatial queries.

Model Name	Primary Fields & Schema Types	Key Indexes & Relations
User	name, email, passwordHash, role (citizen / municipal_staff / administrator), phone, ward, greenPoints, avatar	unique(email), index(role, ward)
Complaint	complaintId, title, description, imageUrl, location (GeoJSON Point), ward, landmark, aiCategory, aiSubtype, aiConfidence, aiPriorityScore, severity, status, sla (dueAt, breached), priorityHistory	2dsphere(location.coordinates), index(status), index(sla.dueAt), ref(reportedBy)
AIAnalysis	complaintId, modelUsed, detections (classId, label, confidence, box), coveragePercent, segregationStream, rawResponseJson	ref(complaintId), index(modelUsed)
Assignment	complaintId, assignedSquadId, squadName, assignedBy, notes, specialInstructions, ppeChecklist, dispatchedAt, completedAt	ref(complaintId), ref(assignedSquadId), index(status)
Verification	complaintId, beforeImageUrl, afterImageUrl, submittedByStaff, citizenStatus (pending / approved / rejected), citizenNotes, verifiedAt	ref(complaintId), index(citizenStatus)
Hotspot	hotspotId, title, centerCoordinates (GeoJSON), radiusMeters, complaintCount, recurringSeverity, recommendedAction	2dsphere(centerCoordinates), index(recurringSeverity)
Reward	userId, pointsAwarded, transactionType (report_approved / cleanup_verified / voucher_redeemed), referenceComplaintId, balanceAfter	ref(userId), index(createdAt)
Notification	recipientId, title, message, type (complaint_update / sla_escalation / reward), isRead, deepLinkUrl	ref(recipientId), index(isRead, createdAt)
SystemSetting	slaThresholds (criticalHours, mediumHours, normalHours), duplicateRadiusMeters, autoEscalationEnabled	singleton configuration document
AuditLog	eventType, entityId, performedBy, oldState, newState, timestamp, ipAddress	index(eventType, entityId, timestamp)

7. Regulatory Compliance, Environmental Impact & Future Roadmap

1. CPCB & WHO Biomedical Waste Rules Compliance: India's Biomedical Waste Management Rules strictly mandate the segregated disposal of clinical sharps to prevent occupational bloodborne pathogens (HIV, Hepatitis B/C) among sanitation workers. CleanTrack automatically detects sharps and routes them to color-coded puncture-proof disposal channels.

2. Municipal Fleet & Fuel Optimization: 50-meter spatial duplicate clustering prevents redundant squad mobilizations, reducing municipal diesel consumption, vehicle wear, and carbon emissions.

3. Future Scalability Roadmap:

- *IoT Sensor Integration:* Ultrasonic bin fill-level telemetry syncing with CleanTrack dispatch routes.
- *Autonomous Drone Patrols:* Automated aerial surveys of open dump grounds, railway corridors, and riverbanks.
- *Multi-Lingual Voice Reporting:* Speech-to-text reporting in 12 Indian regional languages for universal civic accessibility.